Global CO2 Emissions

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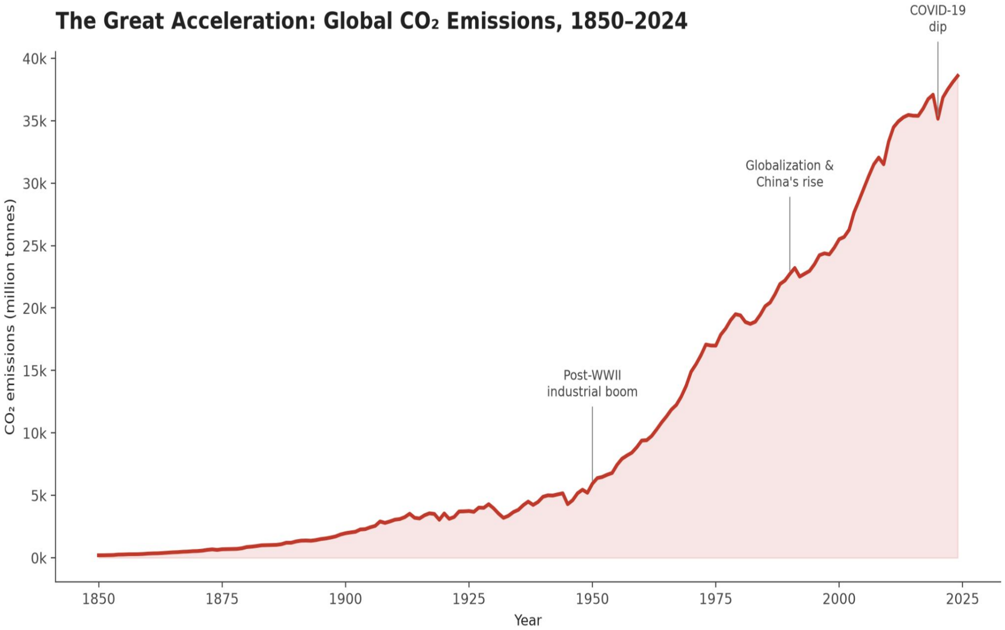
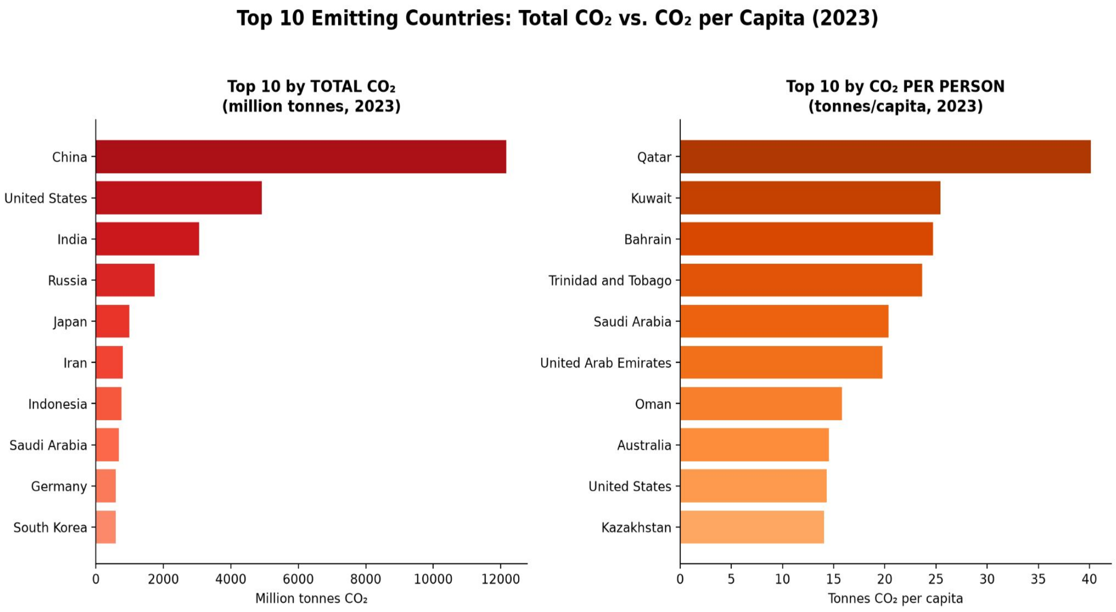
The Global Trend



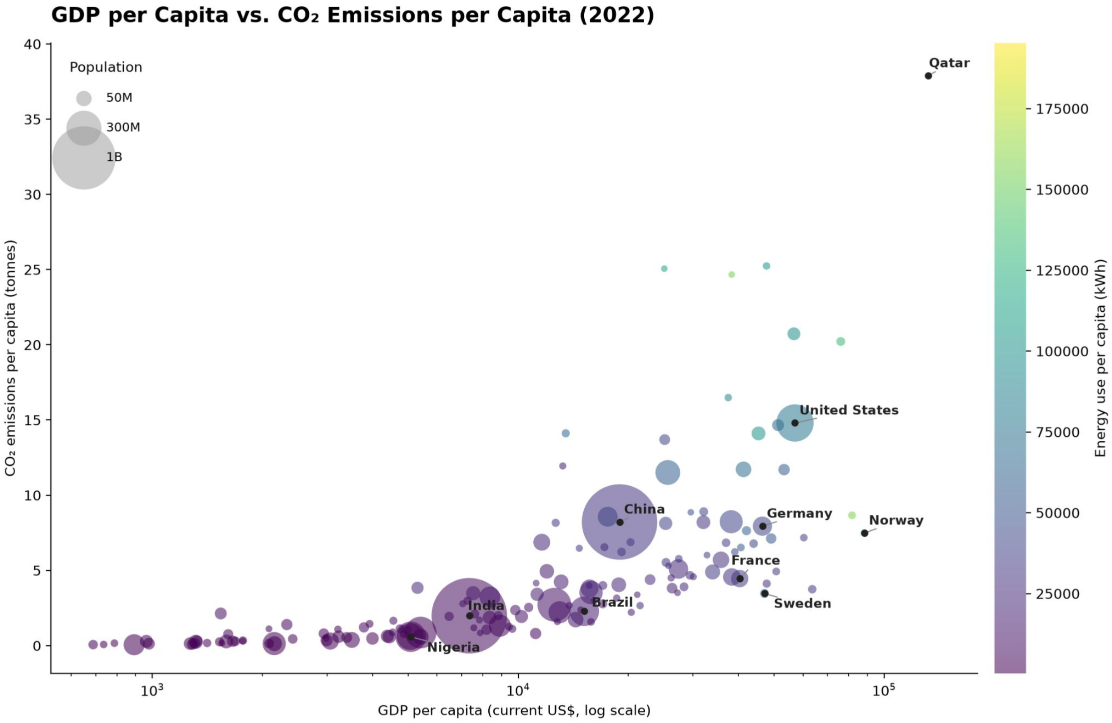
- Global CO2 emissions grew from near-zero in 1850 to ~38 billion tonnes in 2024
- Three inflection points: post-WWII industrial boom, 1990s globalization/China's rise, 2020 COVID-19 dip
- Growth has been continuous no sustained plateau yet visible in the data

Who Emits the Most



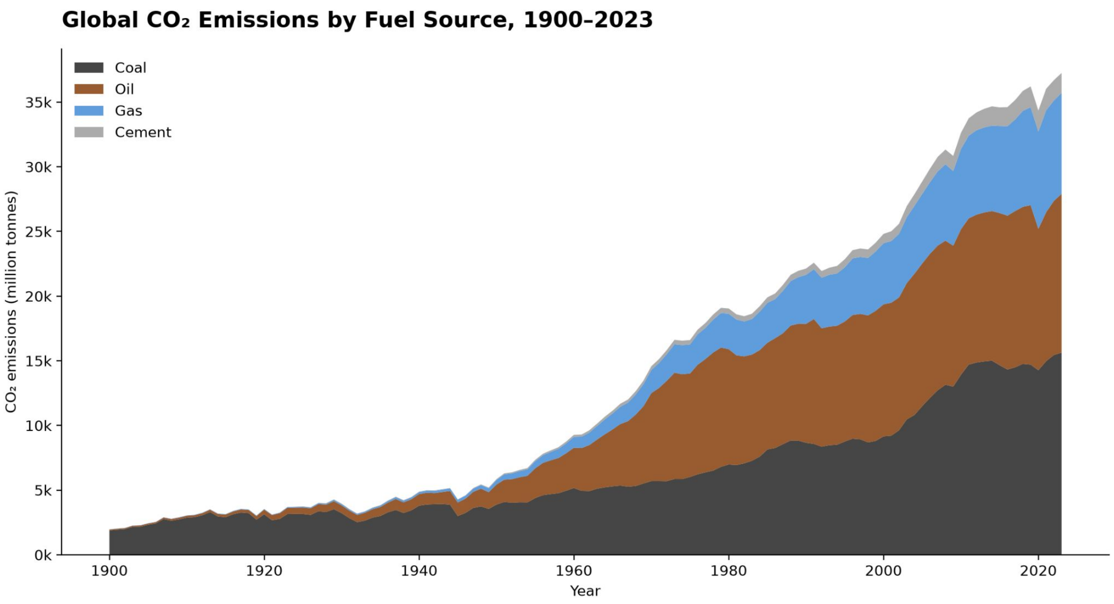
- China, the US, and India lead in total emissions
- A completely different set Qatar, Kuwait, Bahrain, Trinidad and Tobago leads per person
- Total and per-capita rankings tell two different stories: population size matters as much as fuel intensity

Wealth, Energy and Emissions



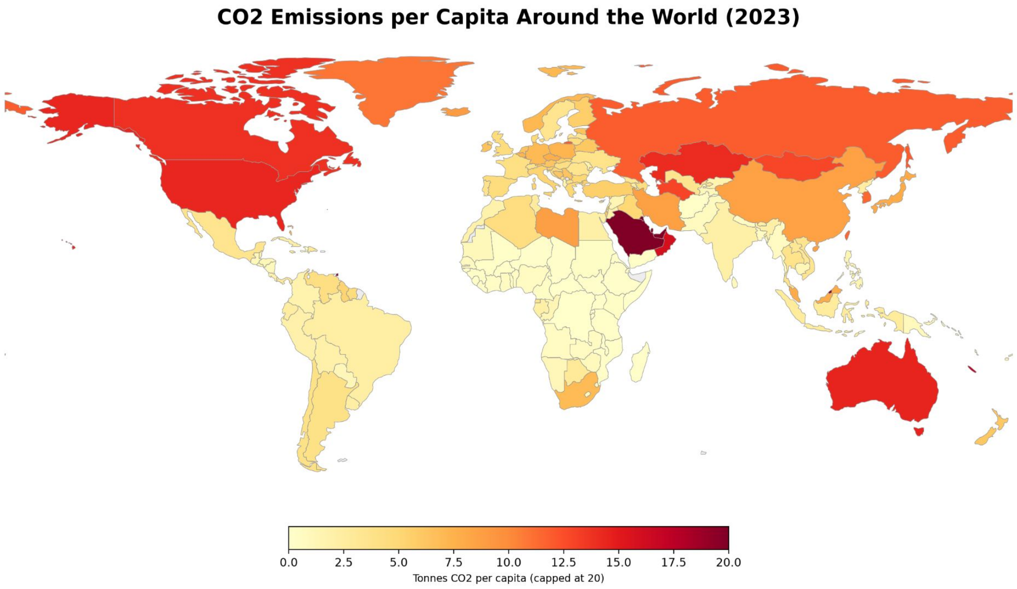
- GDP per capita and energy use per capita are both very strongly correlated with CO2 per capita
- Wealth and emissions are linked but not identical Norway, Sweden, France emit far less than peers at similar income
- Differences likely trace back to energy mix like nuclear, and hydropower rather than wealth alone

The Changing Fuel Mix



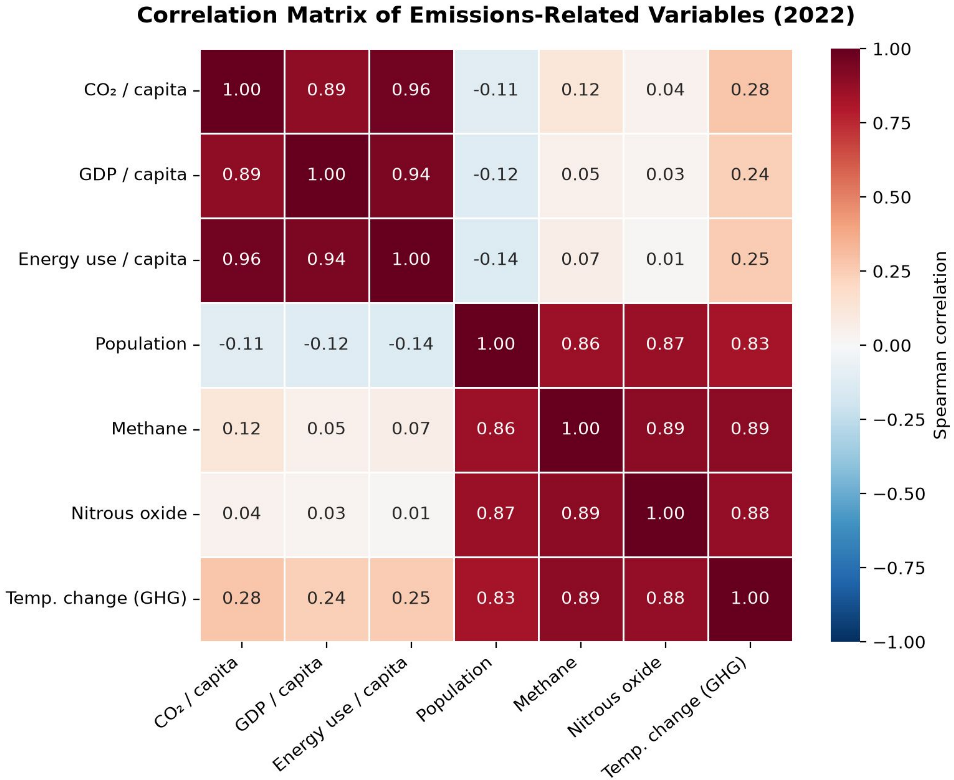
- Coal historically dominated the global fuel mix
- Oil and gas have grown rapidly since the 1950s 60s and now make up the majority of emissions
- The composition of emissions has shifted even as the total has kept climbing

Geographic Pattern



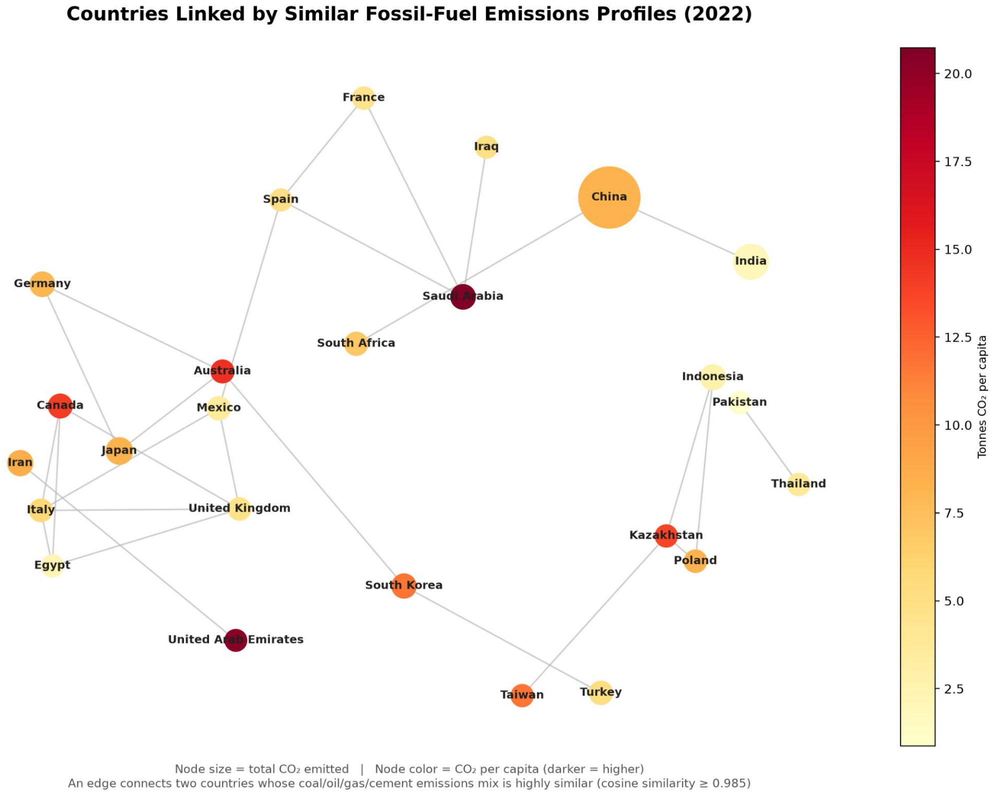
- North America, Australia, and the Gulf states show the highest per capita emissions
- Most of Africa and South Asia remain far below the global average per person
- Geography alone doesn't explain it lines up closely with wealth and fossil-fuel export economies

What Correlates with Emissions



- CO2 per capita correlates strongly with GDP per capita (~0.89) and energy use per capita (~0.96)
- Population size is essentially uncorrelated with per capita emissions
- Population is strongly correlated with total methane and nitrous oxide output instead

Country Clusters



- Countries naturally cluster by how they generate energy, not just by income
- Gulf oil exporting states form one cluster; coal reliant economies form another
- Diversified fuel economies like Germany, Japan, UK, Canada form a third, more balanced cluster

Conclusion

- Emissions growth has been continuous since industrialization only the COVID 19 shock briefly interrupted it
- Who is responsible for this depends on the metric and total points to large economies, per capita points to small wealthyand fuel exporting states
- Wealth and energy use are the strongest predictors of emissions intensity, but fuel mix shifts outcomes even among similarly wealthy countries
- Effective climate policy likely needs to address both total volume and percapita intensity, not a single biggest culprit narrative.